

Mouse Cranial Suture RNA Velocity Analysis

Osteogenic cell differentiation and dedifferentiation trajectories in E17.5 coronal suture

Data: GSE163693 | Farmer et al. 2021, Nature Communications | scVelo + CellRank

OG1

OG2

OG3

OG4

Analysis Overview

Farmer et al. 2021 | GSE163693 | E17.5 mouse coronal suture

1 Data Download

3 BAM files (SRR13288596–98) via aria2c
E17.5 coronal suture, 3 batches

2 velocity

Spliced / Unspliced count generation
Gencode vM25 annotation (chr-prefix removed)

3 scVelo – Full dataset

5,407 cells × 904 genes
Dynamical model, Leiden clustering (19 clusters)

4 OG1–4 Subset

1,772 cells identified by metadata labels
(GSE163693_E15_17_composite_metadata.csv.gz)

5 CellRank (GPCCA)

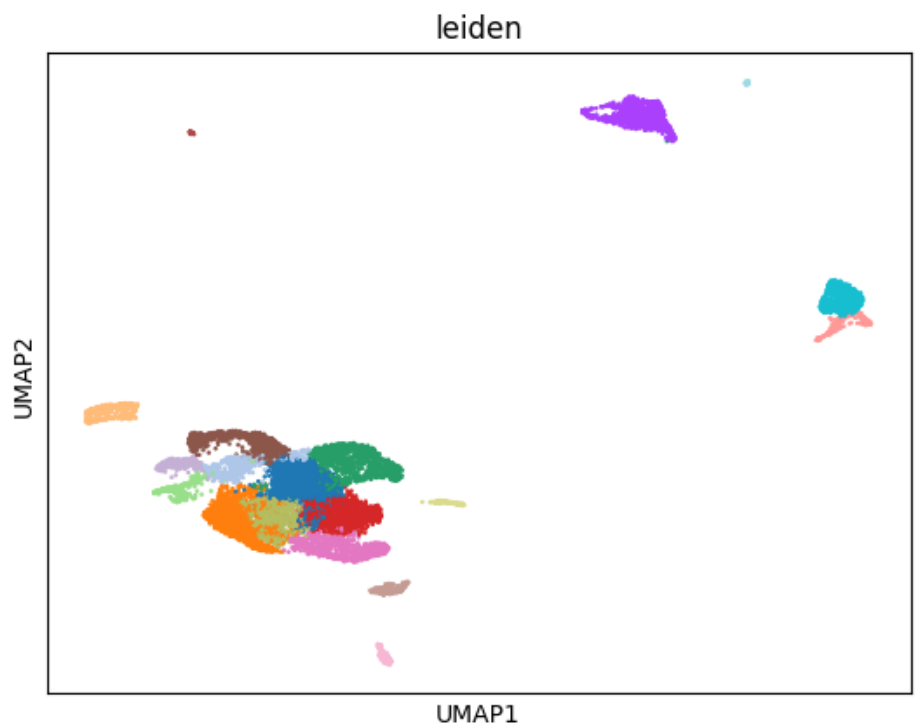
Bidirectional trajectory analysis
Terminal: OG4 + OG1 | Initial: OG3

6 Local Velocity

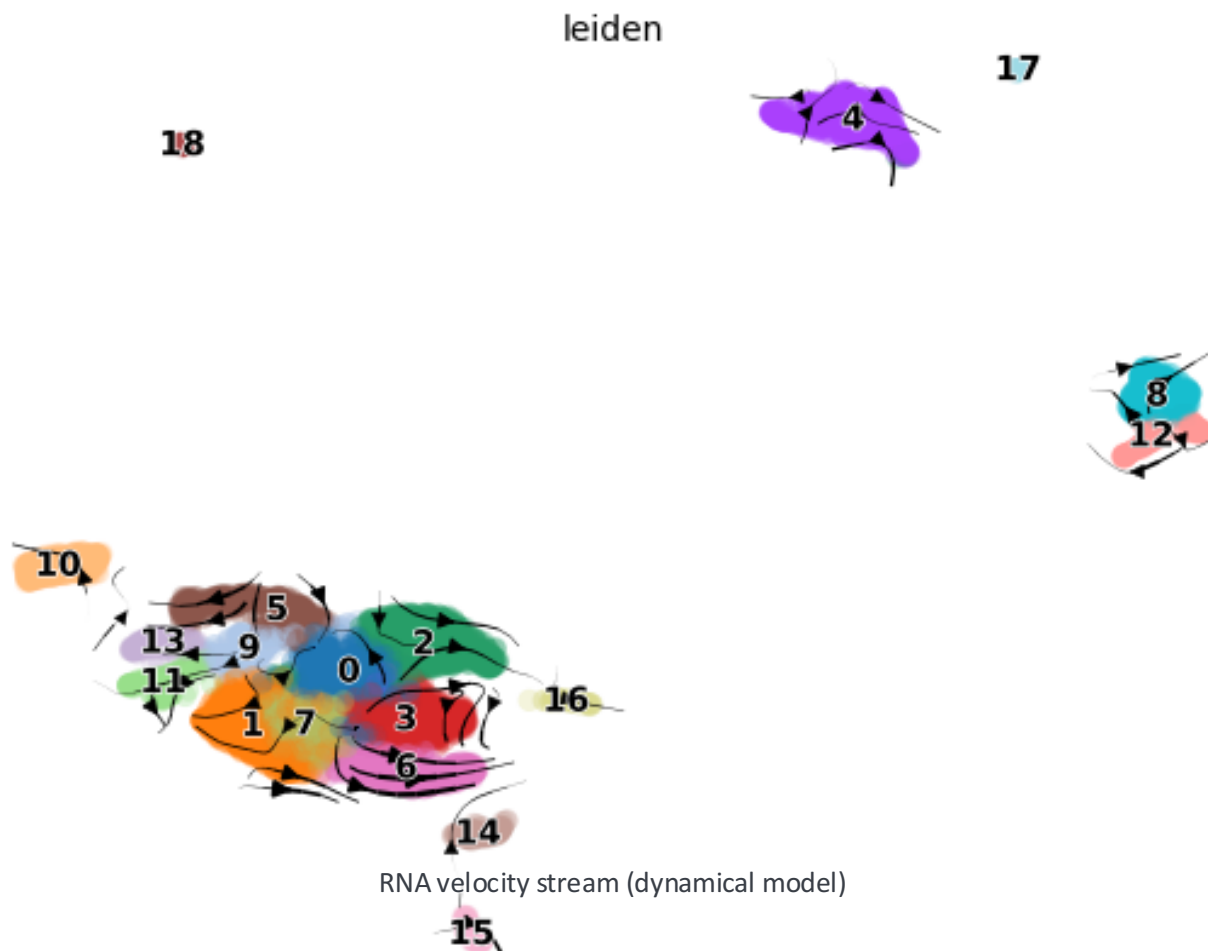
Velocity confidence ≥ 0.77 for all clusters
Phase portraits, PCA-space velocity

Full Dataset: UMAP and RNA Velocity

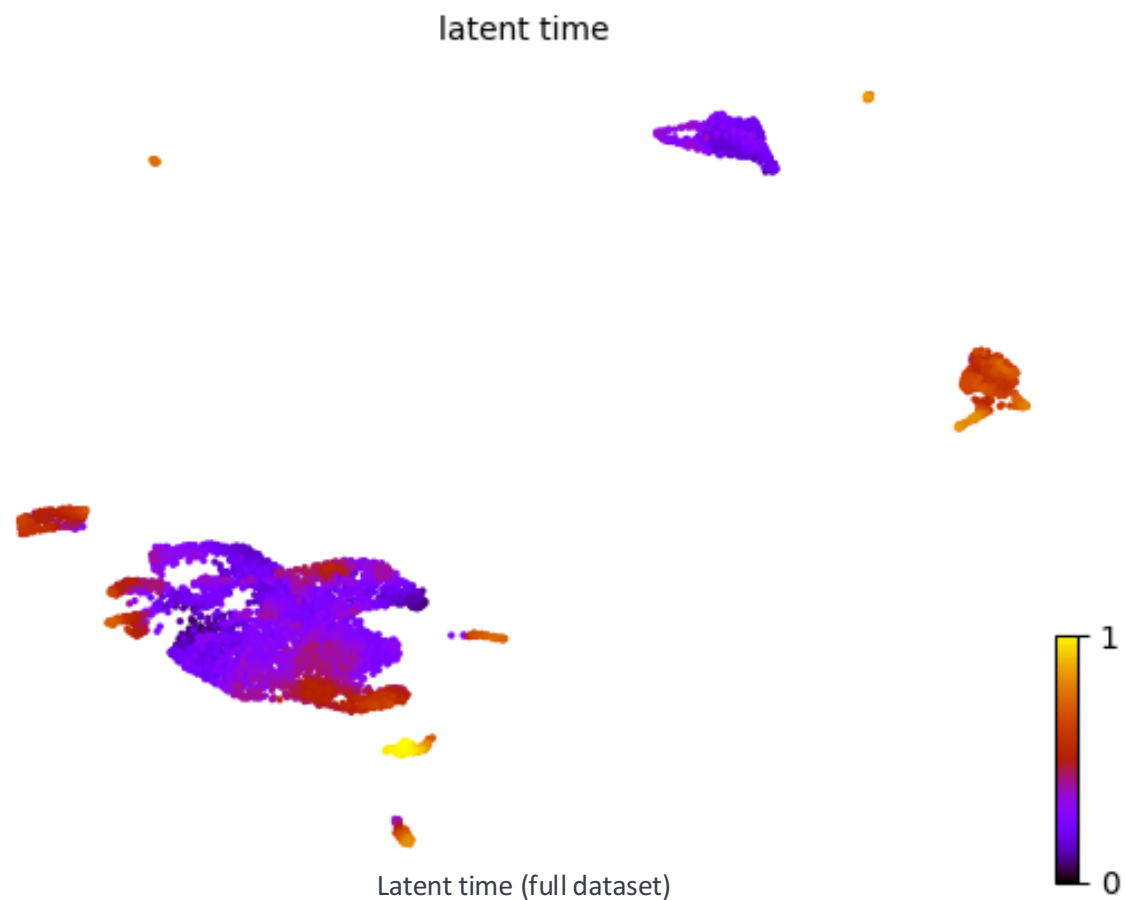
5,407 cells | Leiden clustering (19 clusters) | Dynamical model



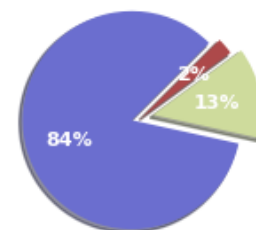
Leiden clusters (19 groups)



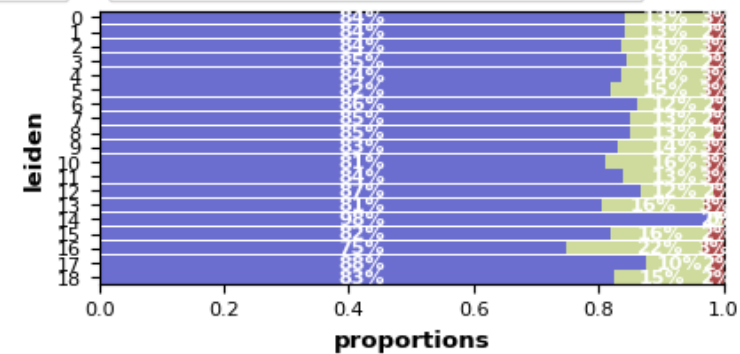
Full Dataset: Latent Time and Splicing Dynamics



spliced unspliced ambiguous



spliced unspliced ambiguous

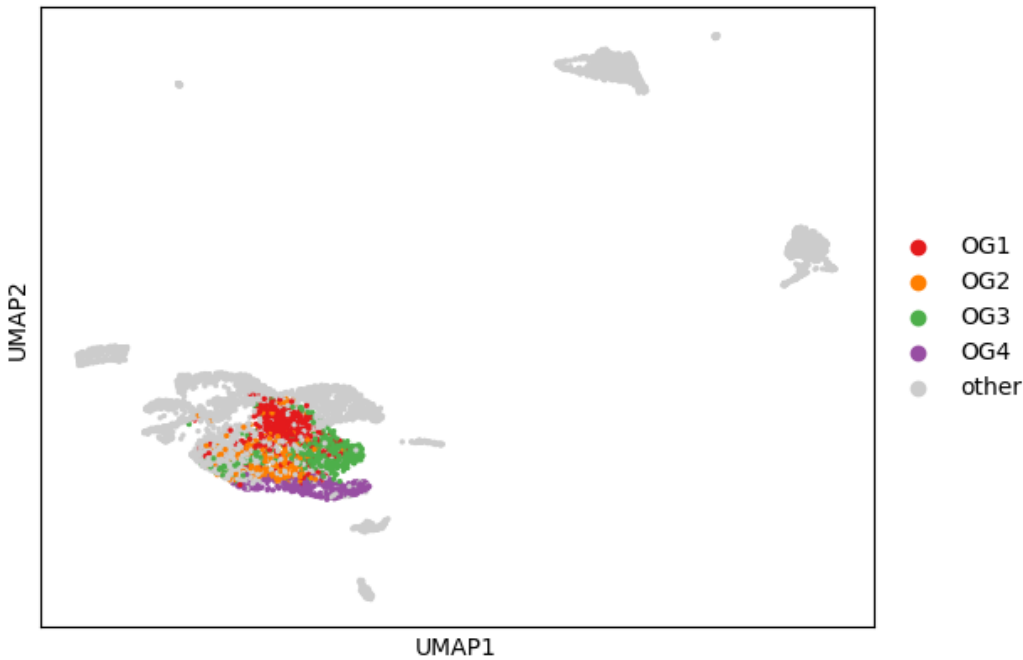


Spliced / Unspliced proportions per cluster

OG1–4 Cluster Definition

Farmer et al. 2021 | Barcode-level identification using metadata labels

OG1-4 location in full UMAP



OG1–4 location in full UMAP

Cluster	Cells	Cell type	Markers
OG1	603 cells	Osteoprogenitors (earliest)	Erg, Pthlh, Six2
OG2	548 cells	Suture-resident pre-osteoblasts	Lef1, Inhba
OG3	315 cells	Periosteal pre-osteoblasts	Mmp13, Podnl1
OG4	306 cells	Mature osteoblasts	Ifitm5, Dmp1, Sost

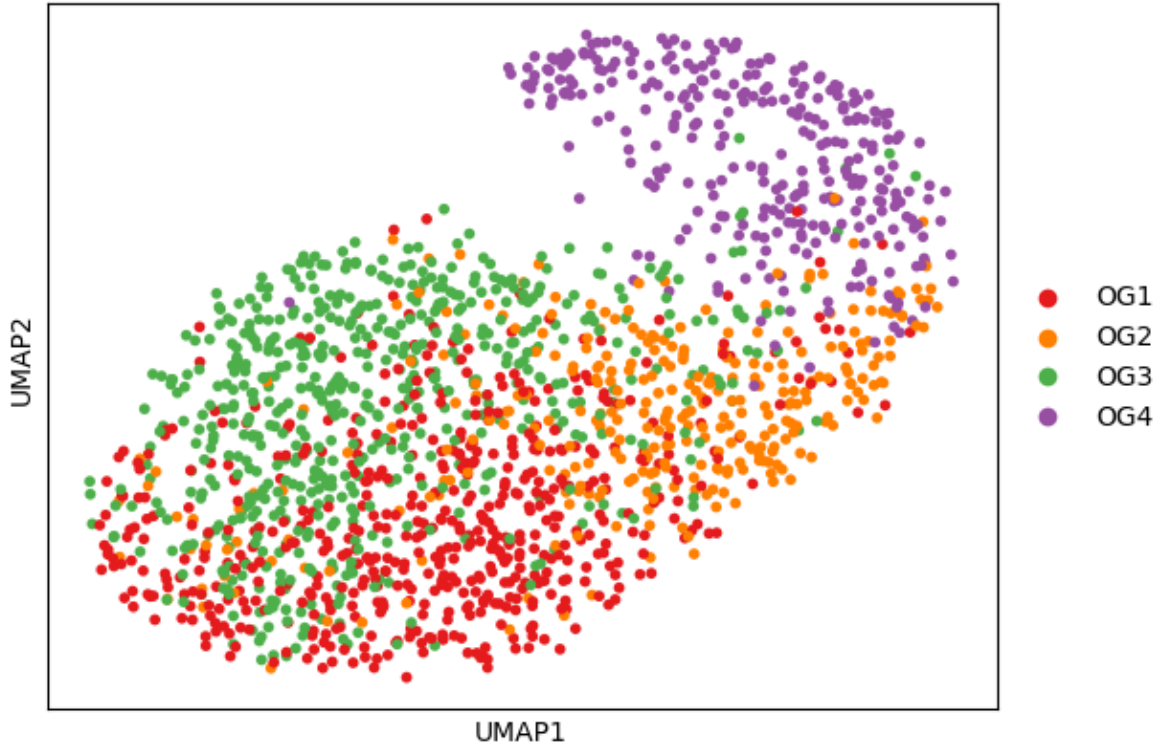
Total: 1,772 E17.5 OG cells

Barcodes matched to GSE163693_E15_17_composite_metadata.csv.gz

OG1–4 Subset: UMAP and RNA Velocity

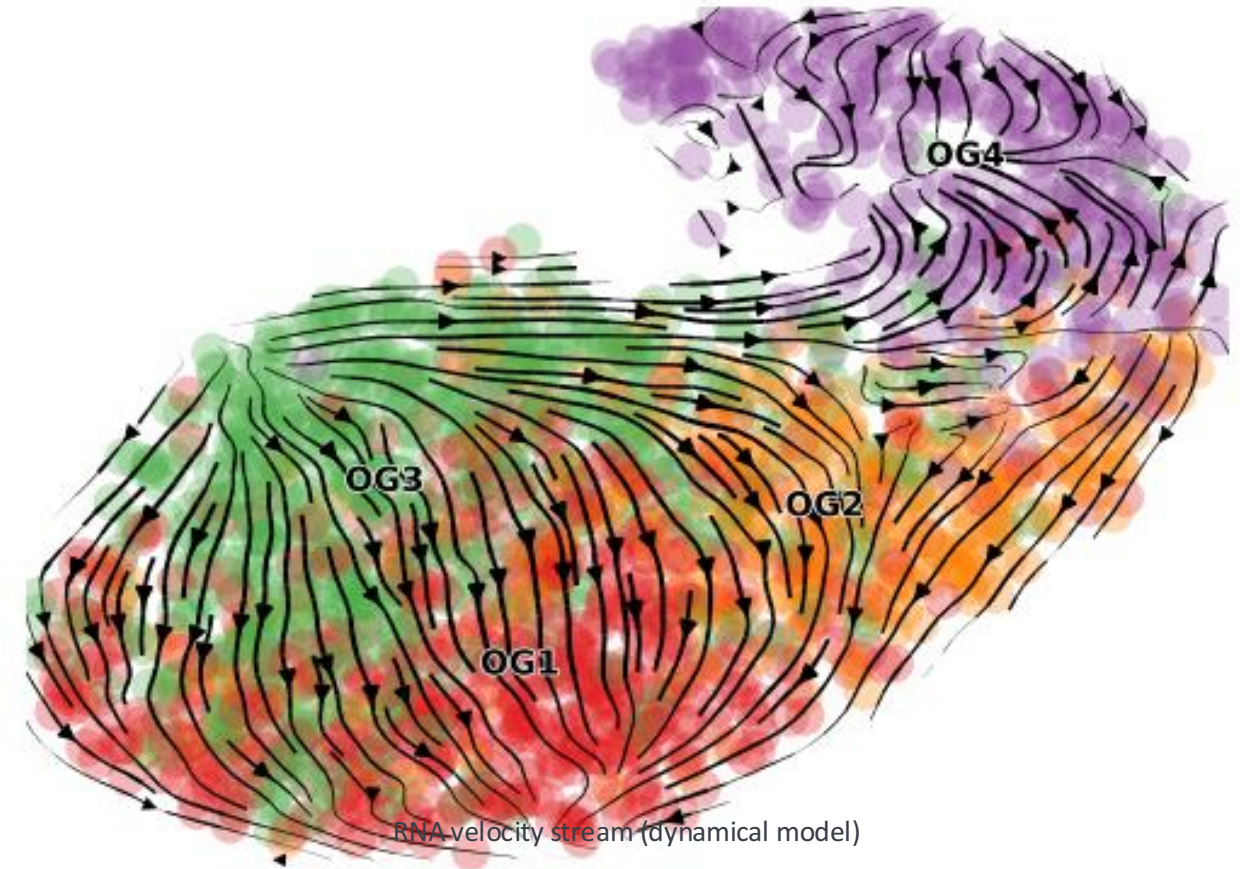
1,772 cells | Re-computed neighbors, UMAP, and velocity | Dynamical model

OG1-4 subset UMAP



OG1–4 subset UMAP

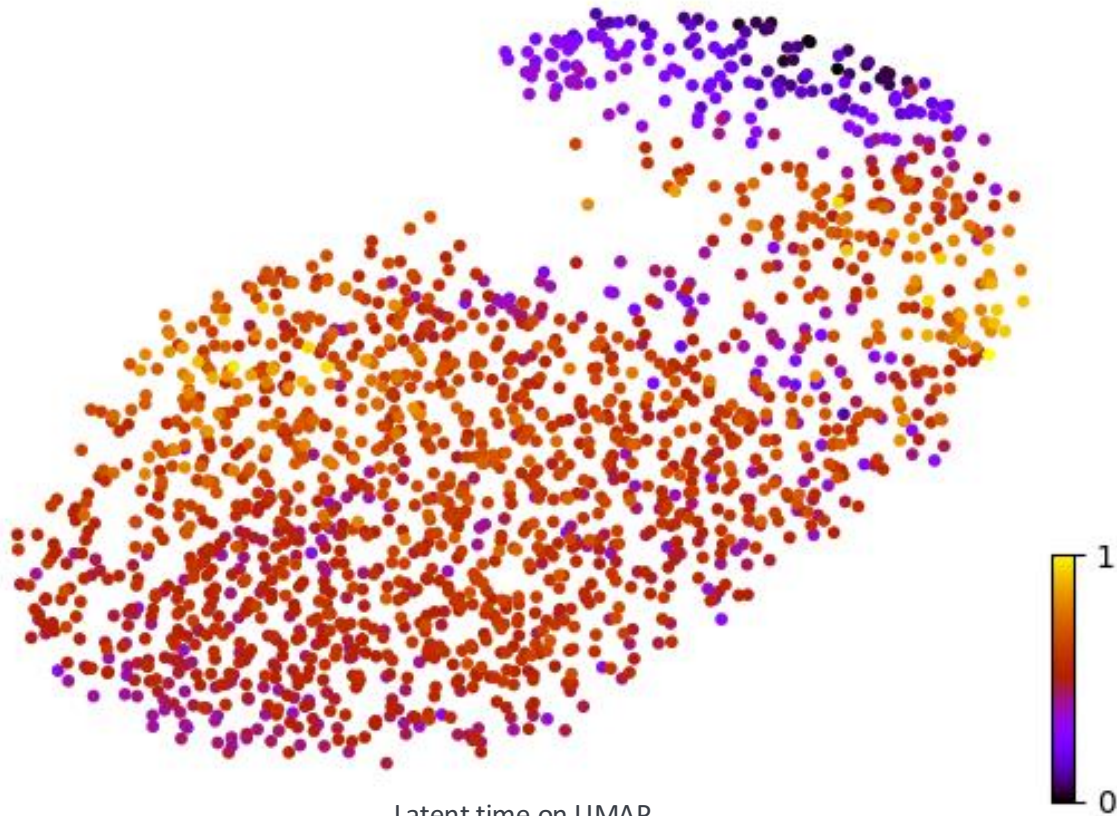
OG1-4 RNA velocity stream



OG1-4 Subset: Latent Time

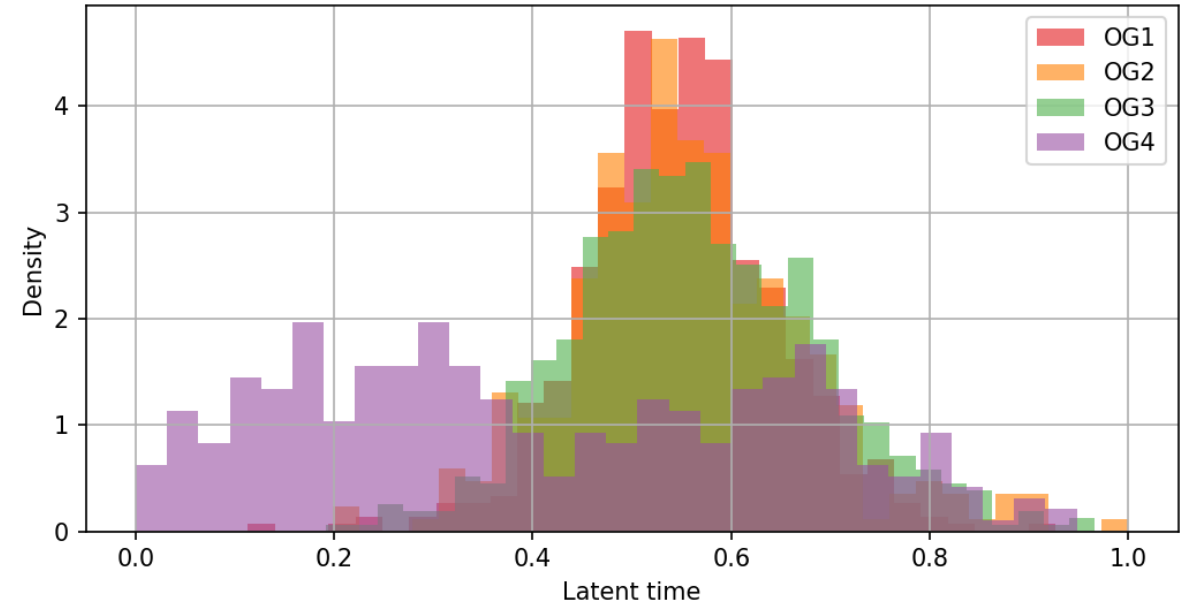
Differentiation pseudotime estimated by scVelo dynamical model

OG1-4 Latent time



Latent time on UMAP

Latent time distribution per OG cluster

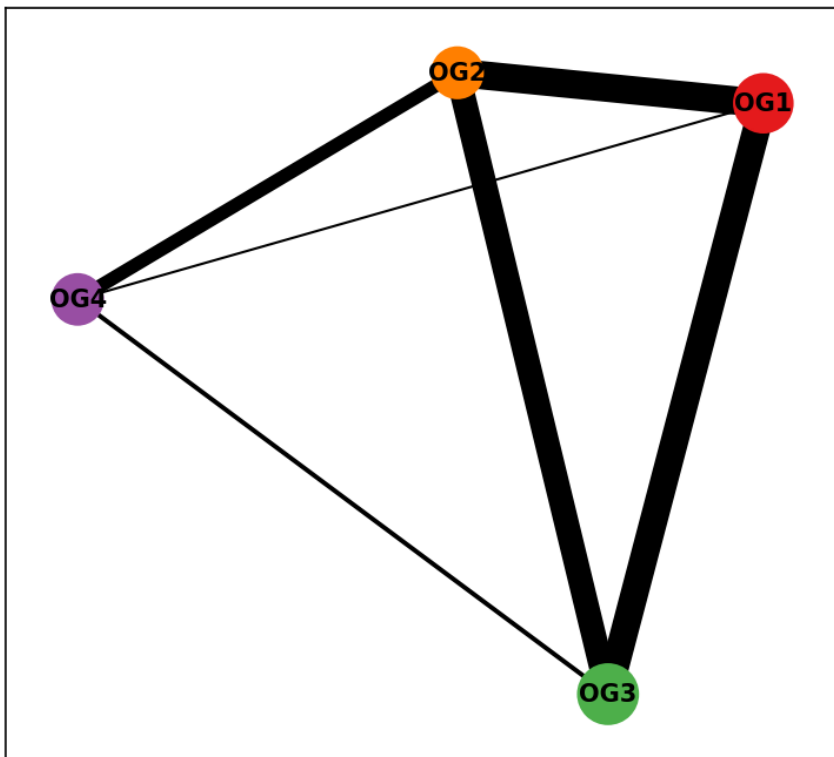


Latent time distribution per OG cluster

PAGA Topology and CellRank Analysis

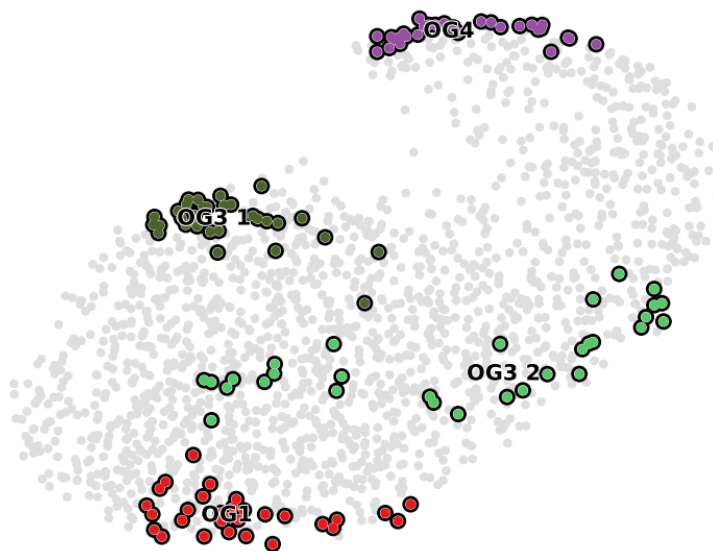
Bidirectional analysis allowing dedifferentiation (OG4 → OG2)

PAGA: OG1-4 topology
(edge width = transition strength)



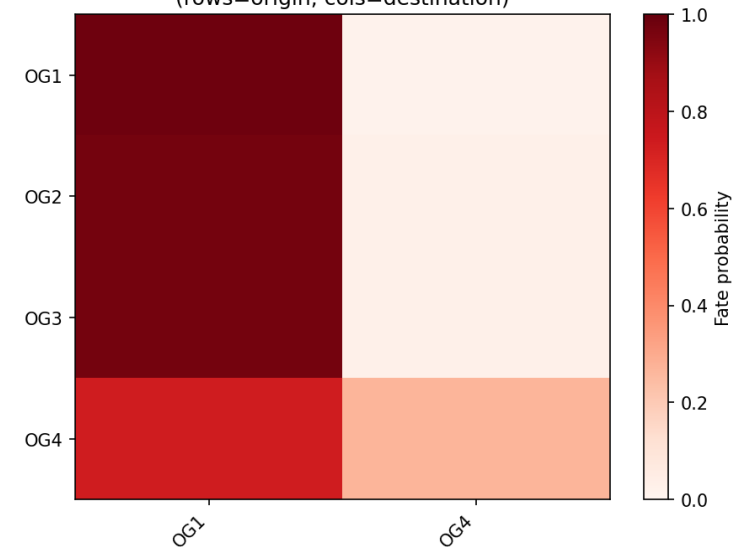
PAGA graph
(edge width = transition strength)

CellRank macrostates



GPCCA macrostates

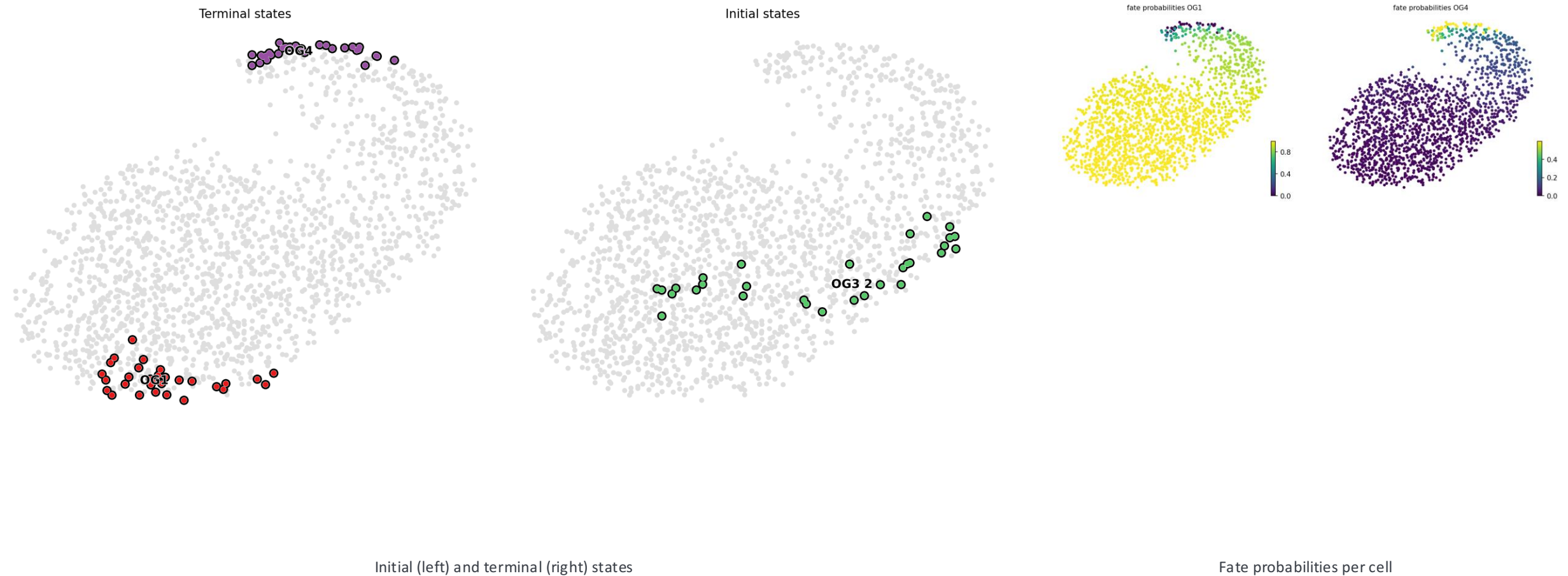
Fate probability per OG cluster
(rows=origin, cols=destination)



Fate probability heatmap

CellRank: Initial / Terminal States and Fate Probabilities

GPCCA identified OG1 as both initial and terminal state → dedifferentiation signal

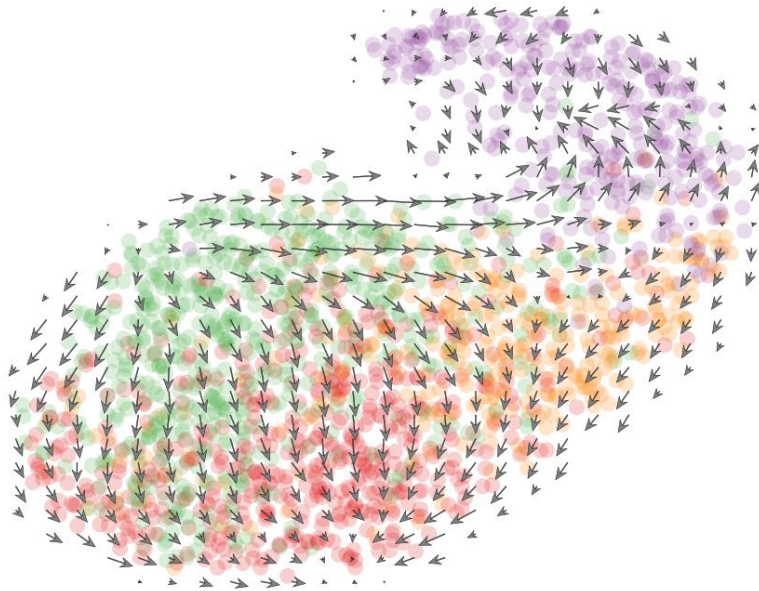


Key finding: OG4 cells show 73% fate probability toward OG1 direction, suggesting a dedifferentiation pathway from mature osteoblasts to progenitors.

Local Velocity Analysis

Grid arrows and phase portraits to distinguish true loop from UMAP projection artifact

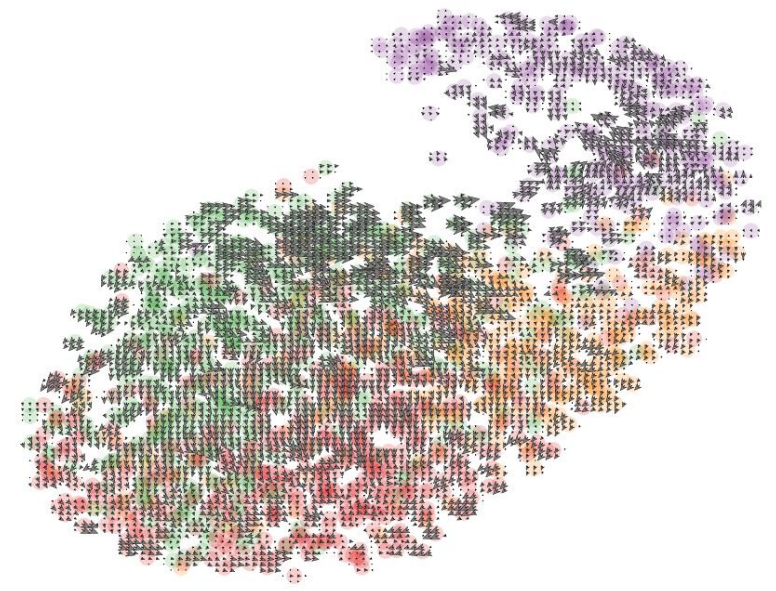
Grid velocity (density=0.5)



Grid velocity (density=1.0)



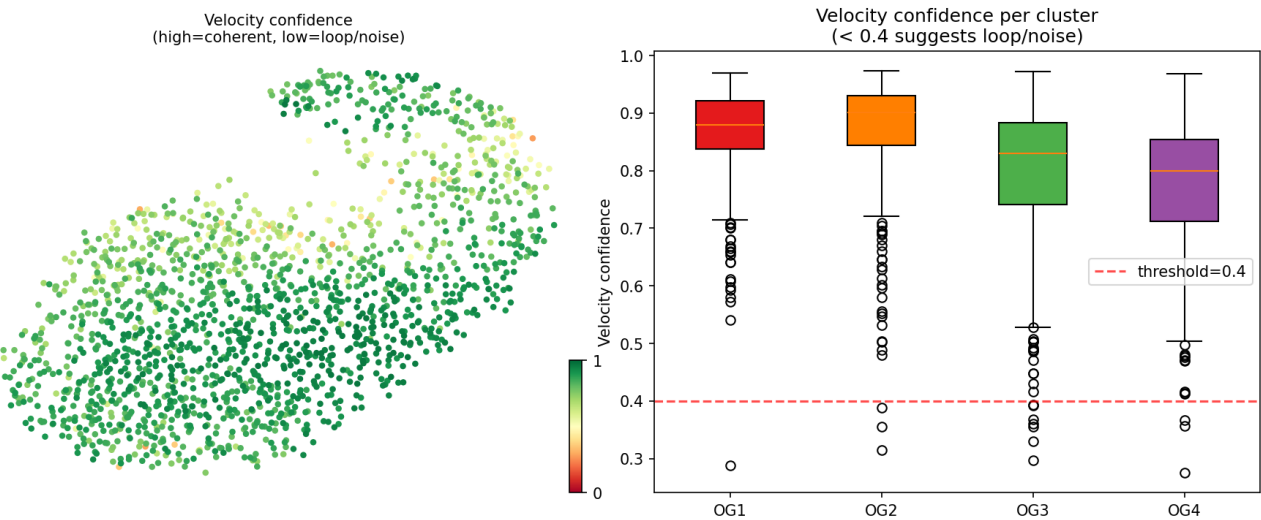
Grid velocity (density=2.0)



Grid velocity arrows at different densities (density = 0.5 / 1.0 / 2.0)
Locally coherent arrows indicate directional flow, not a true cycle.

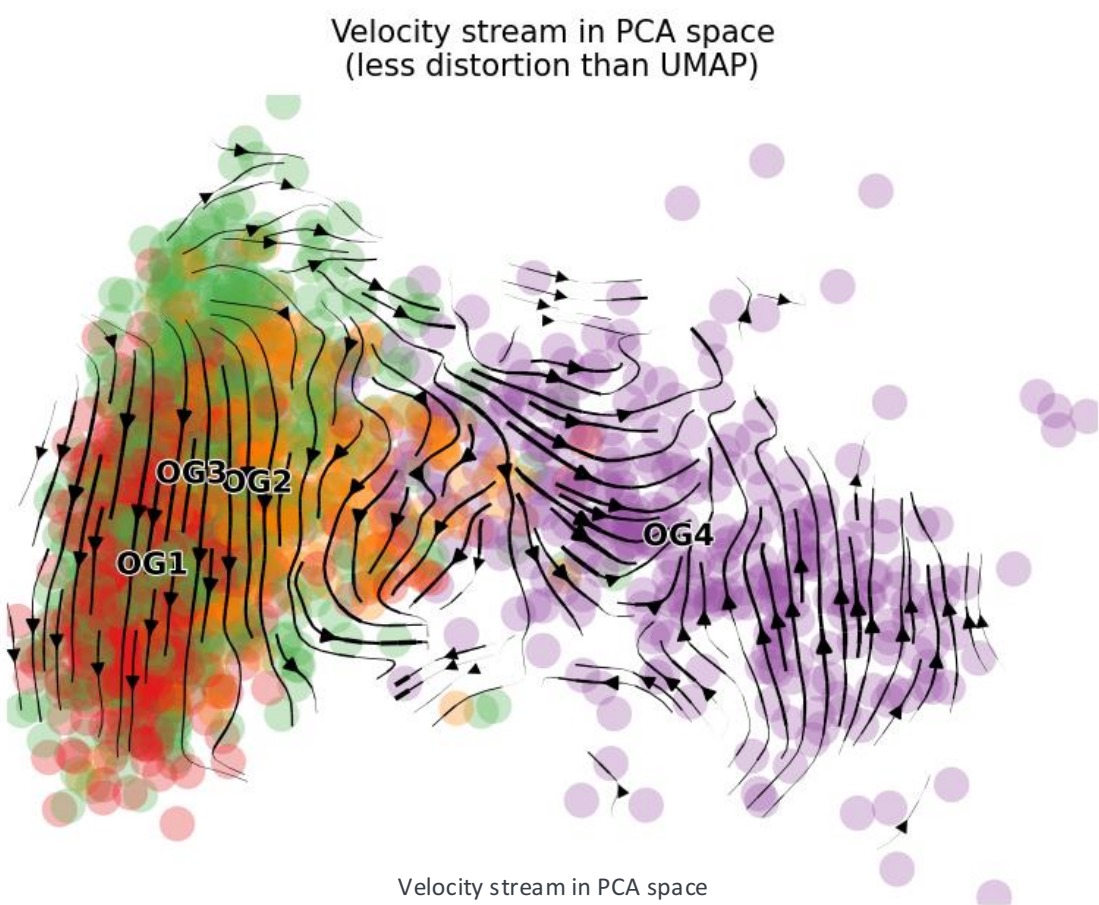
Velocity Confidence and PCA-Space Velocity

Confidence ≥ 0.77 for all clusters | PCA reduces UMAP distortion



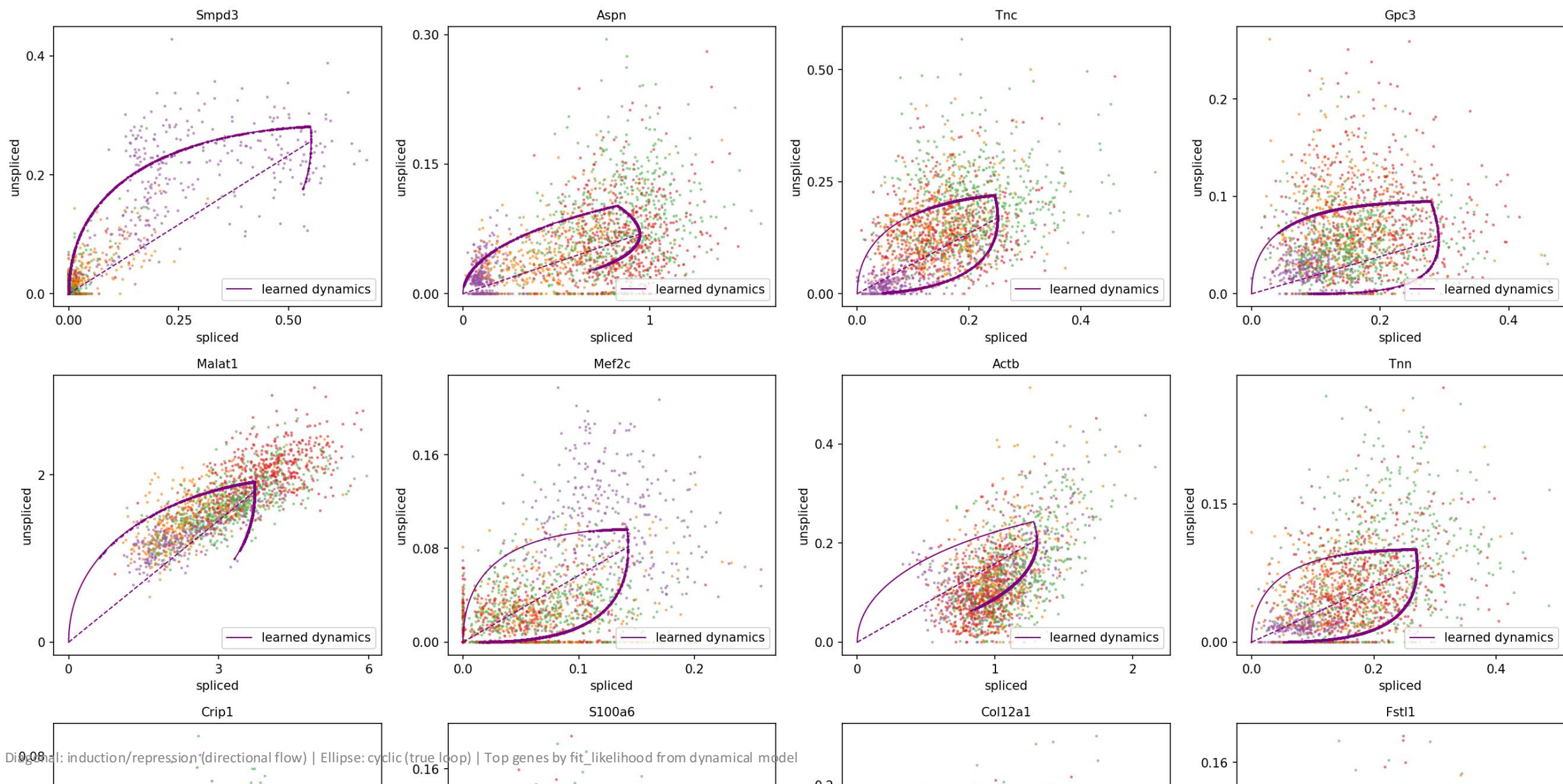
Velocity confidence (left: UMAP, right: boxplot per cluster)

Cluster	Mean	Median
OG1	0.868	0.879
OG2	0.865	0.902
OG3	0.801	0.829
OG4	0.768	0.799



Phase Portraits: Spliced vs Unspliced

Diagonal distribution indicates monotonic kinetics (not a true cycle)



Summary

Differentiation trajectory

OG1 (progenitors) → OG2/OG3 (pre-osteoblasts) → OG4 (mature osteoblasts) confirmed by RNA velocity in E17.5 coronal suture.

Bifurcation

PAGA reveals two branches from OG1: suture-resident (OG2, edge=0.77) and periosteal (OG3, edge=0.75), both converging on OG4.

Dedifferentiation signal

CellRank GPCCA identifies OG1 as a terminal state. OG4 cells show 73% fate probability toward OG1, suggesting dedifferentiation from mature osteoblasts to progenitors.

Loop is a projection artifact

Velocity confidence ≥ 0.77 for all clusters. Phase portraits show diagonal (monotonic) rather than elliptic (cyclic) distribution. The loop in UMAP reflects a Y-shaped bifurcation compressed in 2D.